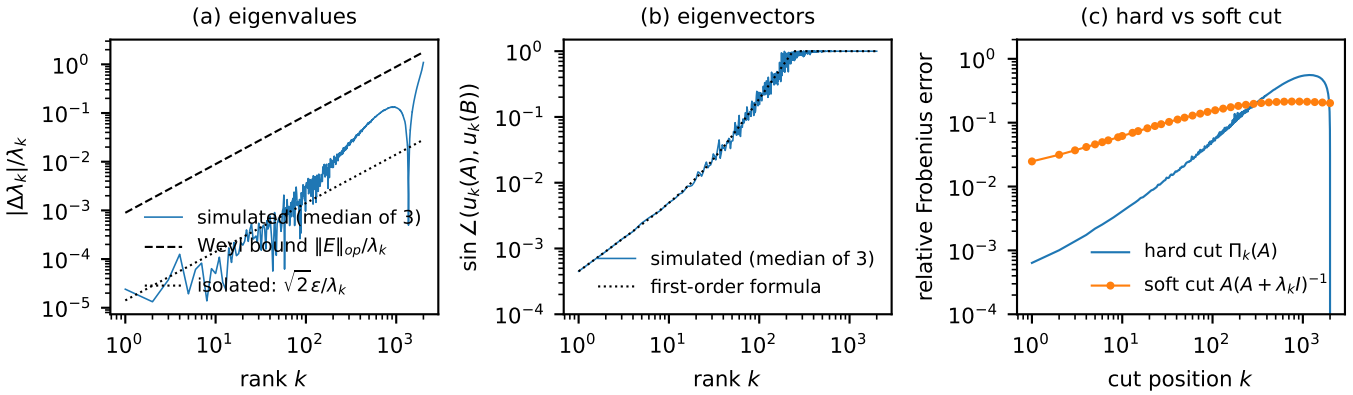
Pure law $\alpha = 1$, $D = 2000$, symmetric Gaussian perturbation with entry scale $\varepsilon = 10^{-5}$ ($\|E\|_{op} \approx 8.9e-04$)
(a) eigenvalues
$|\Delta\lambda_k|/\lambda_k$
simulated (median of 3)
Weyl bound $\|E\|_{op}/\lambda_k$
isolated: $\sqrt{2}\,\varepsilon/\lambda_k$
rank k
(b) eigenvectors
$\sin \angle(u_k(A), u_k(B))$
simulated (median of 3)
first-order formula
rank k
(c) hard vs soft cut
relative Frobenius error
hard cut $\Pi_k(A)$
soft cut $A(A+\lambda_k I)^{-1}$
cut position k